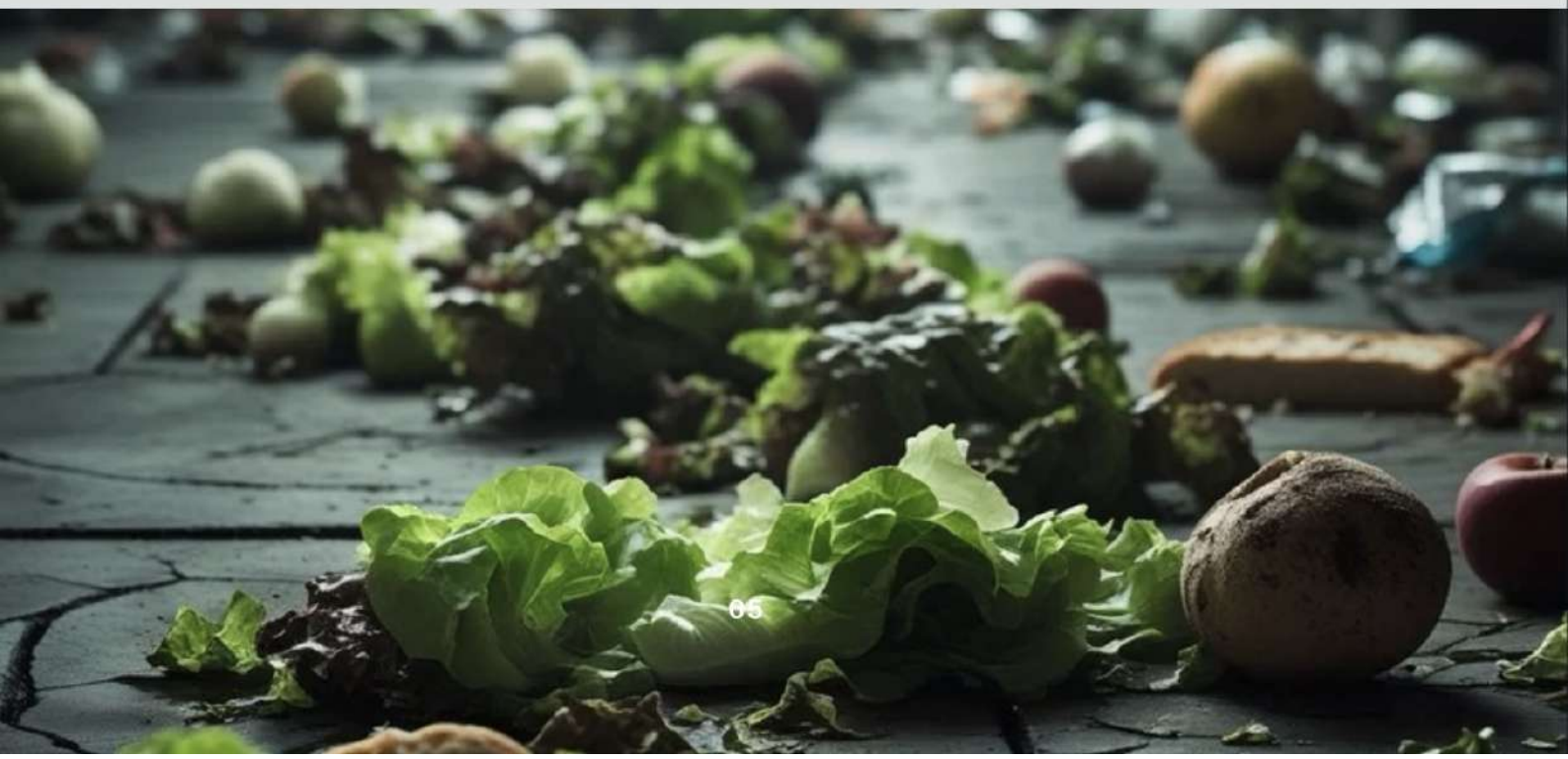
# FOOD WASTAGE IN HOSTEL MESSES



# WHAT EXACTLY IS A PROTOTYPE?

A prototype is an early sample, model, or version of a product, system, or concept that's created to test and demonstrate its functionality, design, and potential. It's a working representation that helps identify strengths, weaknesses, and areas for improvement before building the final version.

# THE SHOCKING SCALE OF HOSTEL FOOD WASTE (AND WHY IT MATTERS)

## The Problem

Hostle and meses generate significant food waste everyday, contributing to environmental damage, resource depletion, and financial loss

## The Solution

Our prototype aims to reduce food waste by leveraging data-driven insight and engaging students

# INTRODUCING "WASTE-NOT": OUR PROTOTYPE APP EXPLAINED

## Intuitive Interface

Easy-to-use interface designed for students.

## Real-Time Data

Tracks food waste in real-time, providing valuable insights.

## Interactive Feedback

Engages students with polls and surveys to understand preferences

# TECHNICAL DEEP DIVE: DATA COLLECTION, ANALYSIS, AND ALGORITHM

## Data Collection

The app collects real-time data on food waste, including quantity, type, and reasons.

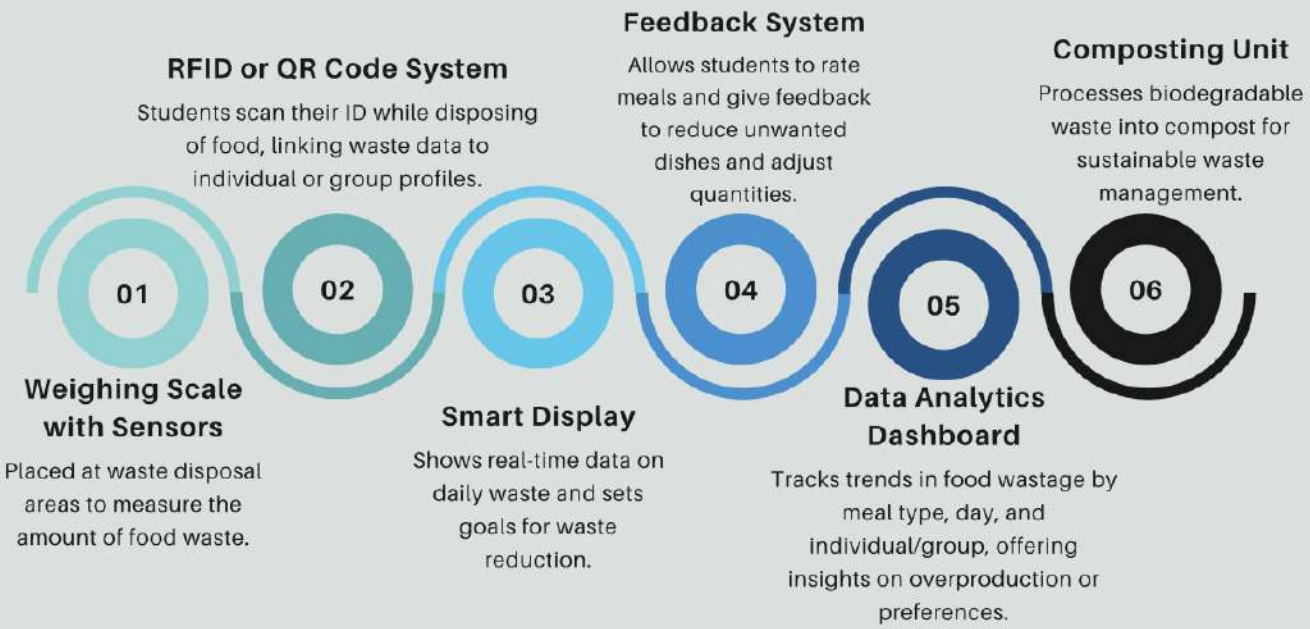
## Data Analysis

Advanced algorithms analyze collected data to identify trends and patterns.

## Predictive Algorithm

The app uses predictive models to forecast future food waste, allowing for proactive adjustments.

# Components


01
Weighing Scale with Sensors
Placed at waste disposal areas to measure the amount of food waste.
02
RFID or QR Code System
Students scan their ID while disposing of food, linking waste data to individual or group profiles.
03
Smart Display
Shows real-time data on daily waste and sets goals for waste reduction.
04
Feedback System
Allows students to rate meals and give feedback to reduce unwanted dishes and adjust quantities.
05
Data Analytics Dashboard
Tracks trends in food wastage by meal type, day, and individual/group, offering insights on overproduction or preferences.
06
Composting Unit
Processes biodegradable waste into compost for sustainable waste management.

Students scan their ID and dispose of leftover food.
The weighing scale measures waste and logs data.
Data analytics identifies recurring patterns of waste and under-consumed or over-consumed dishes.Feedback and pre-booking help adjust the menu and quantities for future meals.
Composting reduces the environmental impact of unavoidable waste.

# FOOD WASTAGE IN HOSTEL MESSES: A DIGITAL SOLUTION

A study by the World Bank estimated that 1/3 of food produced globally ends up as waste. In university hostels, food waste is a significant problem, causing financial strain and environmental damage. This presentation explores the challenges we face and how a food wastage-based app can make a difference.

# BENEFITS OF A FOOD WASTAGE APP

## Reduced Waste

The app allows students to order only what they need, reducing the amount of food wasted by the mess.

## Cost Savings

The app helps optimize food purchasing and preparation, leading to cost savings for the mess and potentially lower food prices for students.

## Sustainability

By reducing food waste, the app promotes a sustainable approach to food consumption, benefiting the environment and the community.

https://docs.google.com/forms/d/1grBCuwWZHOHMNg3fXN6EanXBVsvzbdzFpUD1rzKwn8s/edit?pli=1uu